

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 1

MARK SCHEME

CANDIDATE
NAME

CLASS

BIOLOGY

Paper 2

8875/02

September 2009

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.

Write in dark blue or black pen.

You may use soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **one** question.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
5 or 6	
Total	

This document consists of **11** printed pages and **1** blank page.

[Turn over

Section A

Answer **all** the questions in this section.

- 1 When present in water, carbon dioxide dissolves, forming carbonic acid, which then dissociates immediately to form bicarbonate ions and protons:

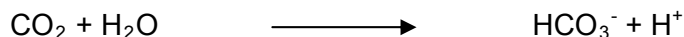


Fig. 1.1 shows the energy changes which take place during this reaction.

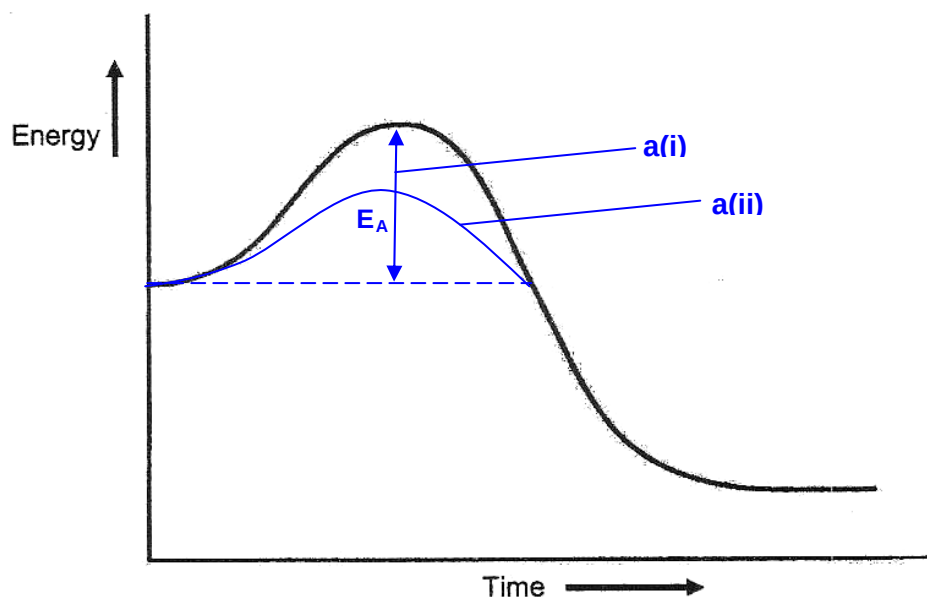


Fig. 1.1

The reaction also occurs in the human blood, where it is catalyzed by the enzyme carbonic anhydrase. The pH of blood is 7.4.

- (a) On Fig. 1.1, sketch and label the energy changes which take place when carbonic anhydrase is present. [1]
- (b) Explain how carbonic anhydrase brings about the energy changes in the enzyme-catalyzed reaction.

Allow close proximity of reactants due to temporary binding of substrates ;

Ensures correct orientation of reactants;

Destabilizing the bonds of reactants as enzymes distort reactants' bonds;

Active site creates a water free zone for non-polar reactants may react more easily;

R-groups of enzymes can change the charge / alter the distribution of electrons on the substrate;

[3]

- (c) **Fig. 1.2** shows the rate of reaction-partial pressure of CO_2 curves for carbonic anhydrase. Graph **A** shows the curve in the presence of acetazolamide, which is an inhibitor to carbonic anhydrase.

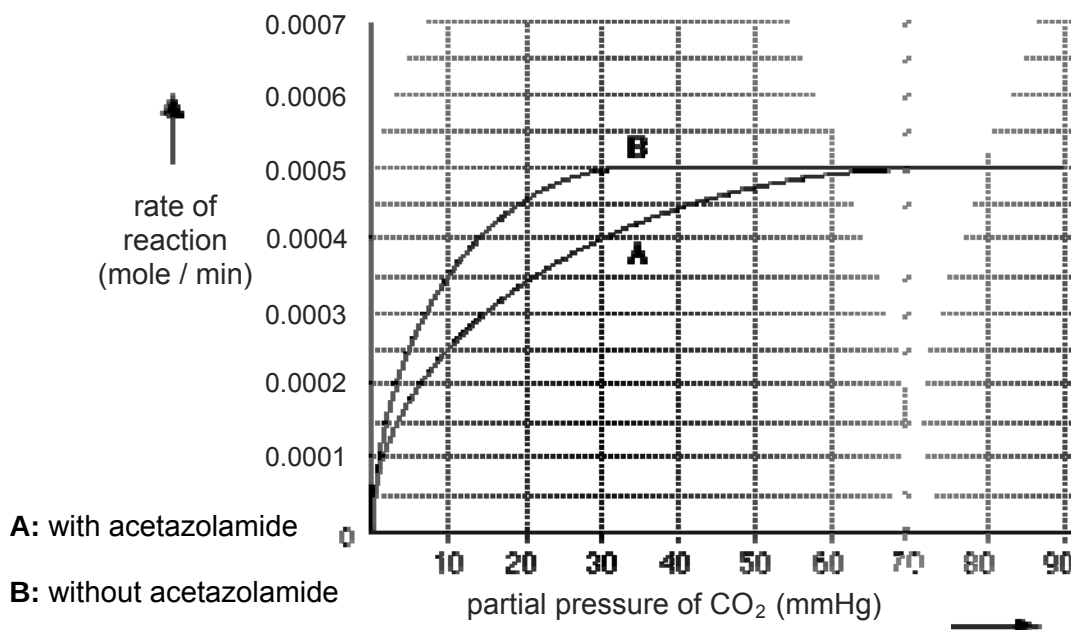


Fig. 1.2

- (i) With reference to the curves in **Fig.1.2**, explain the mechanism by which acetazolamide inhibits carbonic anhydrase.

Competitive Inhibition as V_{max} can be reached at high ppCO_2 ;

Competes with substrate for active site at low ppCO_2 , thus rate of reaction is lower;

Inhibitor is structurally similar to that of substrate molecule / complementary to active site of enzyme;

[3]

- (ii) Suggest why acetazolamide is not broken down by carbonic anhydrase.

Catalytic residues / domains of enzyme unable to recognize / act on bonds of inhibitor;

[1]

[Total: 8]

- 2 **Fig. 2.1** is a diagram showing the sequence of events in the production and secretion of a hormone by a cell.

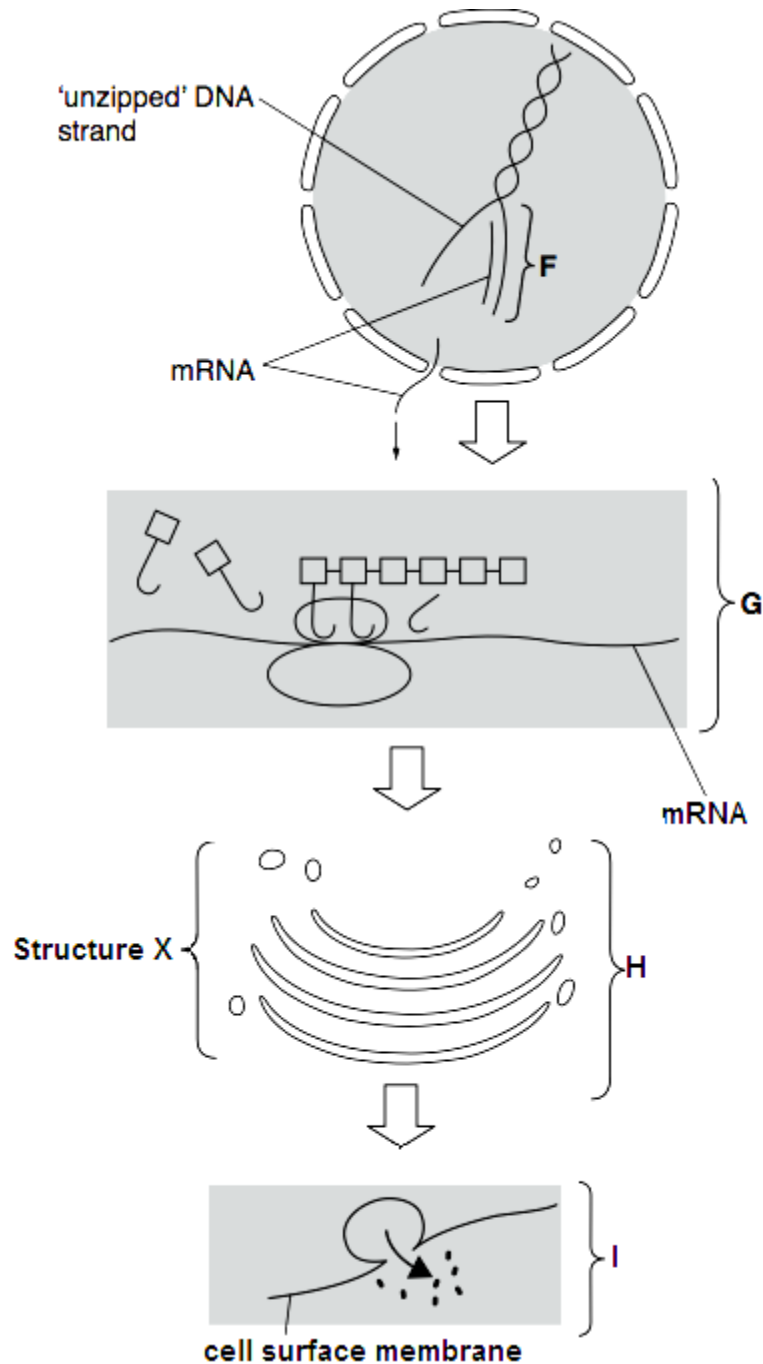


Fig. 2.1

- (a) Name the processes occurring at F, G, H and I.

Any 2 points for 1 mark

F	Transcription
G	Translation
H	Post-translational modification @ any function of GA
I	Exocytosis / secretion

[2]

- (b) Outline the events that occur during F, leading up to G.

During F, mRNA is synthesized by RNA polymerase using DNA as the template;

mRNA undergoes post-transcriptional modification / RNA processing and enter cytoplasm via nuclear pore;

Ribosomal subunits, aminoacyl-tRNA and mRNA associate to carry out translation at G;

[3]

- (c) Describe the function of structure X.

Chemical modification of hormone;

Temporary storage / Packaging hormone into vesicle;

Concentrates hormone molecules into vesicles;

[1]

- (d) Part of the DNA base sequence for this hormone is shown **Fig. 2.2**.

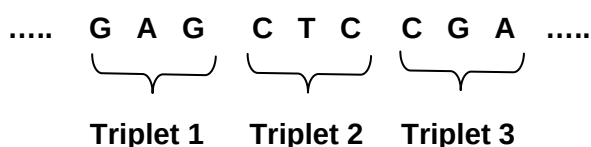


Fig. 2.2

- (i) In **Table 2.1**, state the messenger RNA (mRNA) codon and the transfer RNA (tRNA) anticodon that correspond to each of these DNA triplets. [1]

	Triplet 1	Triplet 2	Triplet 3
mRNA codon	CUC	GAG	GCU
tRNA anticodon	GAG	CUC	CGA

Table 2.1

- (ii) Explain the relationship between the sequence of the DNA triplets and the structure of the hormone.

Sequence of nucleotide on DNA determines sequence of amino acids in polypeptide;

Different R-group interactions result in unique three-dimensional structure of hormone;

[2]

[Total: 9]

- 3 (a) The dog whelk (*Nucella lapillus*) is commonly found on rocky sea shores around the coasts of the British Isles. Considerable variability of its shell thickness and shape occurs on shores exposed to different strengths of wave action, as shown in **Fig. 3.1**.

On sheltered shores they have a high, thick shell and a relatively small aperture for attachment to rocks; on exposed shores the shell is thinner and stubbier, so that they are better able to hold on to the rocks they inhabit when subjected to the pounding of waves. The main predators of dog whelks are crabs, which will be active almost entirely on sheltered shores. Shell thickness is clearly protective here.

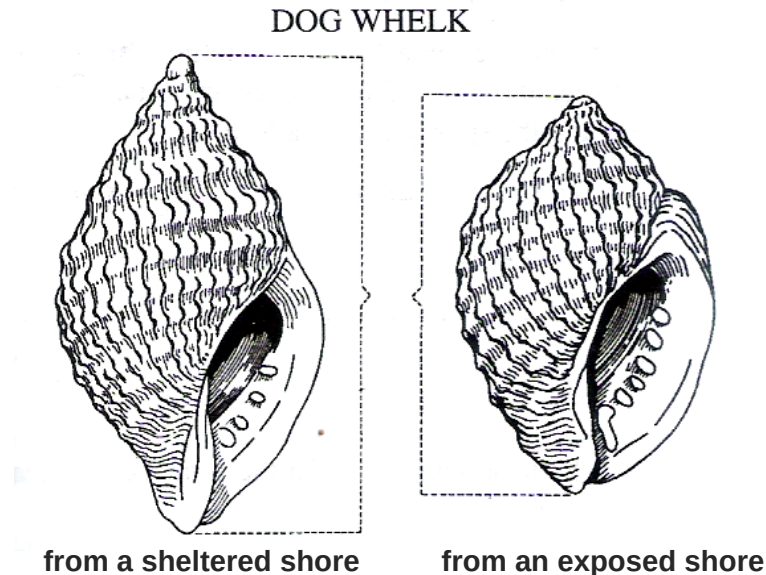


Fig 3.1

Describe the role of natural selection, in maintaining the shell shape and thickness of populations of dog whelks on

- (i) sheltered rocky shores;

Variation in thickness of shells of dog whelks,;

Predation from crab exerts selection pressure on sheltered shores;

Whelks with thick shells are selected for (while thin shells are selected against);

Likes produce like, thus population of thick shelled whelks pass their desirable traits to offsprings (award only once in (i) or (ii));

[3]

- (ii) exposed rocky shores.

Variation in shell shape / aperture size of dog whelks,

Pounding waves act as selection pressure on exposed rocky shores;

Thin and stubby-shelled whelks / whelks with smaller aperture will be selected for as they are not washed away (while thick-shelled whelks / whelks with small aperture for attachment to rocks are selected against);

Likes produce like, thus population of thin shelled whelks pass their desirable traits to offsprings (award only once in (i) or (ii));

[3]

- (b) The two types of dog welks are classified as varieties of the same species.
Describe how a molecular method can be used to show relatedness of these dog welks.

Compare DNA sequence of a common gene / amino acid sequence of a common protein;

High percentage similarity between sequence will indicate that both types are closely related;

[2]

[Total: 8]

- 4 Scientists have recently produced genetically-engineered clover containing a protein high in sulphur-containing amino acids. As proteins in wool are also rich in these amino acids, it is thought that feeding sheep on this new strain of clover (leaves are consumed) will produce an increase in the yield of wool. A DNA fragment was prepared from the sequences shown in **Fig. 4.1** for insertion into the clover.

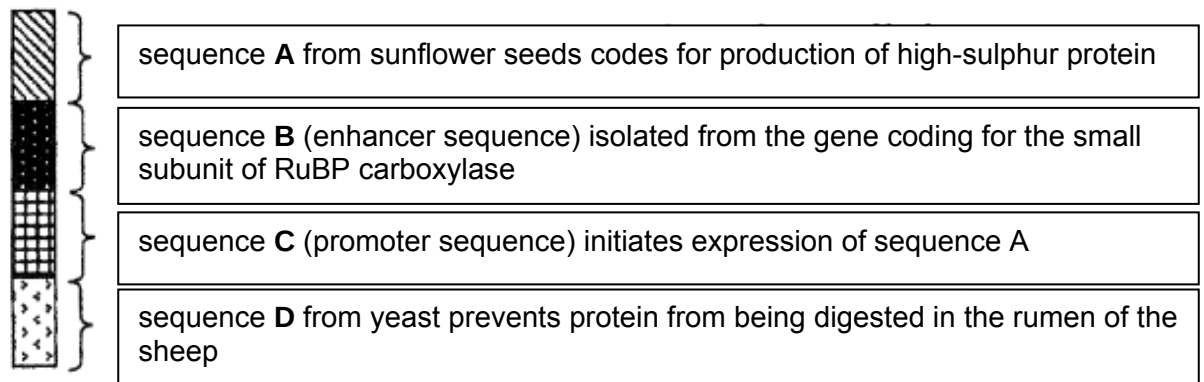


Fig 4.1

- (a) Describe how enzymes are used to construct the DNA fragment shown in **Fig. 4.1**.

Restriction enzymes are used to synthesize complementary sticky ends between each fragment;

DNA ligase is used to form phosphodiester bonds between nucleotides / fragment;

[2]

- (b) Suggest why sequence **B** was included in the DNA introduced into the clover.

Increase the expression of sequence A;

In photosynthesizing cells which are in the leaves of the plant;

[2]

- (c) Fig. 4.2 shows a vector map of the plasmid vector pSING003. The plasmid is used to clone the DNA fragment, which will be inserted at the *EcoRI* site. The plasmid is subsequently introduced into *E. coli* via transformation. After transformation, the *E. coli* cells are then grown on an agar plate.

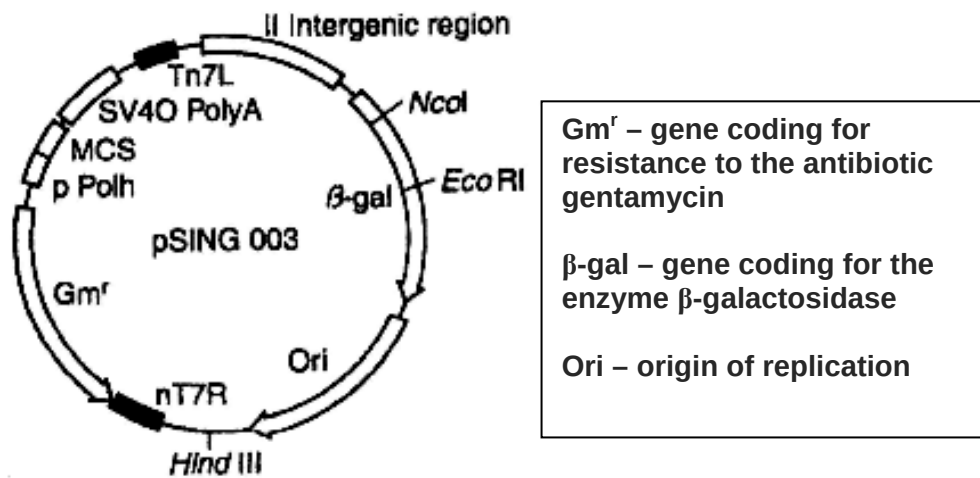


Fig. 4.2

- (i) Describe **two** properties of plasmids that allow them to be used as DNA cloning vectors.

Possess own origin of replication thus can replicate independently of main chromosome;

Possess restriction sites which can be cut by restriction enzymes for insertion of gene-of-interest;

[2]

- (ii) In addition to the nutrients required for growth, name **two** other substances that are added to the agar plate to allow selection of cells successfully transformed with the recombinant plasmid.

1. X-gal;
2. Antibiotic gentamycin;

[2]

- (d) **Fig. 4.3** shows the results of plating the transformed cells onto an agar plate with the appropriate substances.

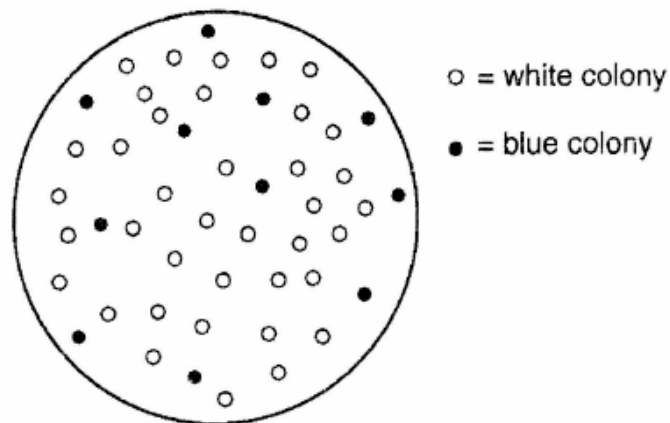


Fig. 4.3

- (i) Explain why some colonies appeared white while others appeared blue.

Intact β -galactosidase gene codes for enzyme which hydrolyzes X-gal (giving rise to blue colour);

Bacteria colonies with reannealed plasmids will appear blue;

Bacteria colonies with recombinant plasmids will appear white;

Due to insertional inactivation which disrupts β -galactosidase gene;

[4]

- (ii) Using **Fig. 4.3**, calculate the percentage of cells into which the recombinant plasmid had been successfully introduced. Show your working in the space provided below.

Percentage of cells = $38/49 \times 100 = 77.6\%$ (correct to 3 sig. fig.);

[1]

- (iii) Suggest why replica plating was not necessary in this experiment.

Replica plating is only needed if the bacteria colonies which possess recombinant or reannealed plasmids are cultured on the same medium and cannot be differentiated;

Bacteria with recombinant plasmids can be differentiated visually by their colours;

[2]

[Total: 15]

Section B

Answer EITHER 5 OR 6 on the Answer Paper provided.

- 5 (a) Describe the role of control elements in transcription. [5]
1. Initiation of transcription is controlled by binding of transcription factors, TATA binding proteins and RNA polymerase;
 2. to promoter, which is a non-coding region found upstream of gene, resulting in basal transcription rate;
 3. Enhancers are non-coding sequences involved in increasing the rate of transcription and are bound by activators
 4. which loops the DNA and stabilizes the transcription initiation complex via protein-protein interaction;
 5. Silencers are non-coding sequences involved in decreasing the rate of transcription and are bound by repressors;
 6. which, block and impede RNA polymerase's progress / recruit enzymes that increase compaction of DNA / recruit proteins which destabilize the transcription initiation complex / prevent binding of activators to enhancer;
- (b) Explain the eukaryotic processing of pre-mRNA. [6]
1. Capping changes 5' end of mRNA to a 3' end, by 5'-5' linkage / with the addition of 7-methylguanosine group;
 2. Protects mRNA from degradation by 5' exonucleases / confers stability / assists in ribosomal binding;
 3. Splicing removes introns and ligates exons to produce a mature mRNA;
 4. Alternative splicing may occur, resulting in different combinations mature mRNA from a gene;
 5. Addition of poly-A tail to 3' end of mRNA;
 6. Allows slower degradation by 3' exonucleases / longer poly(A) tail implies longer half-life of mRNA / facilitates transport of mRNA from nucleus to cytoplasm;

- (c) Compare and contrast between prokaryotic and eukaryotic control of gene expression at the translational and post-translation level. [9]

Feature	Eukaryotic	Prokaryotic
Initiation	1. Initiated by recognition of AUG sites by small ribosomal subunit;	
	2. First amino acid methionine	First amino acid N-formyl-methionine;
Translational repressors	3. Translational repressors bind at 5' UTR to prevent translation initiation;	
Stability of mRNA	4. Eukaryotic mRNA more stable / present in cytoplasm for longer period of time, due to 5' cap and poly A tail	Prokaryotic mRNA has no poly A tail or 5' cap, so less stable / have shorter half-life;
Antisense RNA	5. Antisense RNA produced prevent translation → duplex RNA which ribosome cannot bind / is degraded;	
Biochemical Modification	6. Addition of biochemical groups to proteins → activate protein;	
		7. Biochemical modification not as significant due to lack organelles such as ER & GA;
Protein degradation	8. Unwanted / misfolded proteins can be degraded by proteasomes / lysosomal degradation	Lack lysosomes / proteasomes but possess other mechanism of degrading proteins;
Feedback inhibition	9. Feedback mechanism present to control gene expression;	

[Total: 20]

- 6 (a) Describe how a gene-of-interest can be amplified without the use of cells. [6]

1. Through the use of Polymerase Chain Reaction, which involves semi-conservative DNA replication, where each strand of double helix acts as template;
2. DNA double helix is denatured / undergoes breakage of H bonds / to generate single-stranded DNA at high temperature / 95°C;
3. Specific sequences of DNA can be amplified by the use of specific primers, as primers bind to start and end of desired DNA sequence;
4. Taq polymerase catalyses synthesis of new daughter strand by addition of free nucleotides via complementary base-pairing / A=T, G≡C;
5. Amplification/ chain reaction occurs, whereby products of previous reactions are used as reactant in next cycle / sequential process of denaturation-annealing-elongation repeated many times;
6. Large amounts of DNA accurately produced from minute starting materials and in a short period of time;

- (b) Explain, with specific examples, how genetic engineering can improve the quality and yield of crop plants in solving the demand for food in the world. [9]

Candidate must elaborate on at least 3 specific e.g. which must cover both improvements in 'quality' and 'yield'. If candidate only elaborated in one aspect, maximum of 6 marks will be awarded.

Examples for improvement in yield of crop plants (any 2 e.g.)

Pest-resistant transgenic plants

1. Transgenic plants containing the Bt gene will produce the protoxin which is ingested by the insect; Protoxin is cleaved in the insect gut and the active Bt toxin is released;
2. Protein binds to cell membranes and causes them to be permeable causing gut cells of insect to lyse, eventually killing insects;

Herbicide-resistant transgenic plants

3. Glyphosate is a herbicide used to get rid of weeds which compete with crops for soil nutrients;
4. Inhibits 5-enolpyruvylshikimate-3-phosphate (EPSP) synthase needed for biosynthesis of essential amino acids;
5. Transgenic plants contain mutated gene which codes for mutant EPSP synthase that is not inhibited by glyphosate;

Extension of shelf life of produce

6. Transgenic plants contain antisense polygalacturonase mRNA;
7. Results in the formation of mRNA-antisense RNA hybrid, which prevents the polygalacturonase mRNA from being translated;
8. Enzyme will not be synthesized, thus ensures fruits are not damaged / too ripe / soft when it reaches consumer in market place;

Examples for improvement in quality of crop plants (must have this e.g.)

Improving nutritional value of crops

9. Golden rice is a transgenic plant that three beta-carotene biosynthesis genes;
10. Produces yellow rice grains containing beta-carotene, which is a precursor to vitamin A;
11. Prevent vitamin A deficiency in world's population that depends on rice as a staple food;

(c) Discuss the ethical and social implications of genetically modified organisms. [5]

Social implications (maximum 3 marks)

1. Transfer of antibiotic resistance markers to pathogenic microorganisms which may result in increase in resistance to clinically important antibiotics;
2. Probability of introducing novel allergens as GM foods may contain proteins introduced from sources people are allergic to;
3. Possibility of GM food being toxic or carcinogenic as they may cause over-expression of other proteins;
4. Monopolistic behaviour of biotechnology companies as terminator gene is likely to be inserted into many GMO seeds, causing second generation seeds to be sterile;
5. Scientists have raised concerns about innovations in research that are not shared, raising fears that world food production may be dominated by a few large biotechnology companies;
6. Increasing dependence of developing nations on industrialized nations;
7. Impact on international trade as Europe has been much more hesitant than the United States in accepting GM products in processed food;

Ethical issues (maximum 2 marks)

8. Tampering with nature as it is going against the natural way of life;
9. Lack of mandatory food labelling in some countries;
10. Religious groups are concerned that GM foods might contain genes from animals prohibited by their religion;

[Total: 20]